# Meibomianitis and Associated Skin Disorders

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# SOAP Format

- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = plan
  - For each diagnosis, what is going to be done to help the patient

# Meibomian Gland Dysfunction

- Not considered an infectious entity
- Altered meibomian gland metabolism
- → secretions become thickened and block the gland opening
- → stagnant material becomes growth medium for bacteria
- → bacteria produce lipase and the released fatty acids trigger inflammation

# Meibomian Gland Dysfunction

**Often associated with (acne) rosacea**

- **Etiology**
  - **Chronic inflammatory disorder involving skin and eye**
  - **Sebaceous glands**

- **Epidemiology**
  - **“Celtic”, northern ancestry (genetic component)**
  - **Adults 30-60 years of age**
  - **> 30% have ocular problems**

# Rosacea

- **Redness (erythema) in the “facial flush area”**



# Rosacea

- Papules and pustules in the “flush areas”

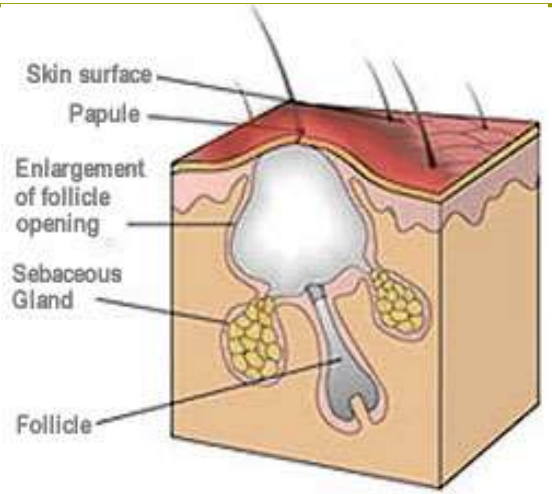




# Rosacea

**Papule**

- **Red and inflamed lesion on the skin's surface**
- **Contains no pus or white head**
- **High break in the follicle wall**
- **Cellular debris and bacteria spill into the dermis.**
- **Often progresses to a pustule**



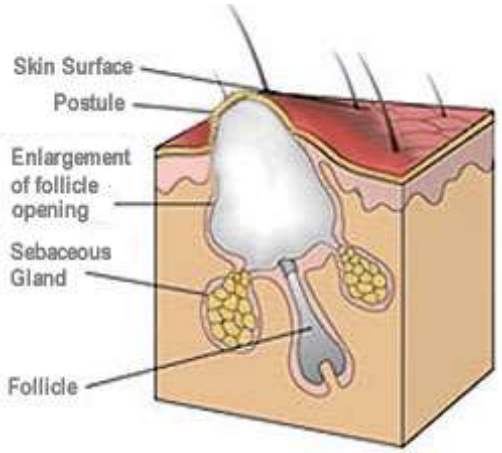

Papule



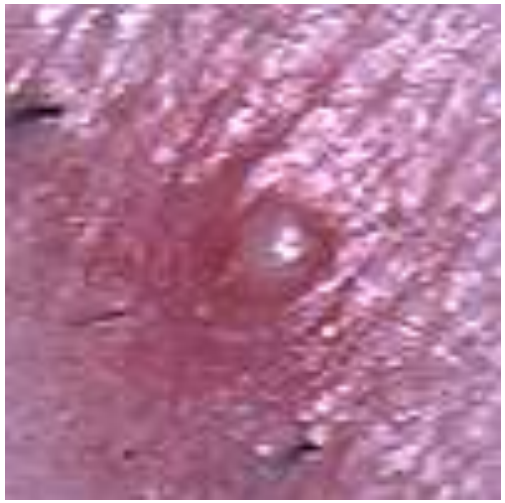
# Rosacea

**Pustule**

- **A typical ‘pimple’**
- **Red and inflamed with a white head**
- **Sometimes brownish core can be seen as a cap on the pus head**
- **Pustules follow papules after white blood cells invade**
- **White cap on a pustule is pus, sebaceous matter and cellular debris**




Pustule



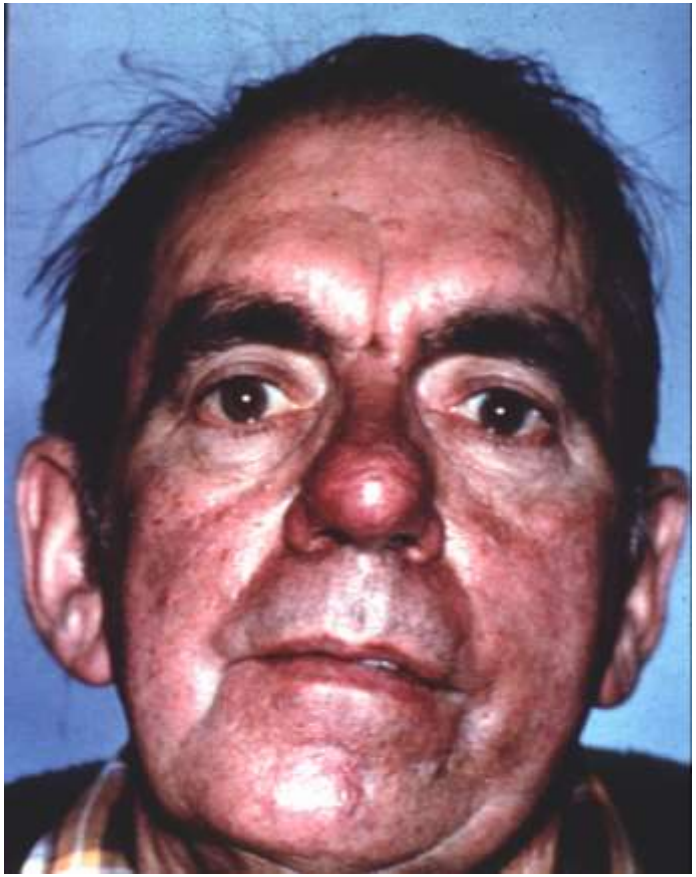
# Rosacea

- **Telangiectasias - visible (dilated) small blood vessels on nose or cheeks**



# Rosacea

- **Rhinophyma - red, bullous nose (sebaceous gland hypertrophy)**



- **W.C. Fields 1880 – 1946**



# Rosacea

## Treatment of facial rosacea

- Doxycycline 100 mg bid x 4-6 weeks
- Topical metrodiazole gel 0.75% bid x 5-8 weeks






# Rosacea

## Treatment of Facial Rosacea



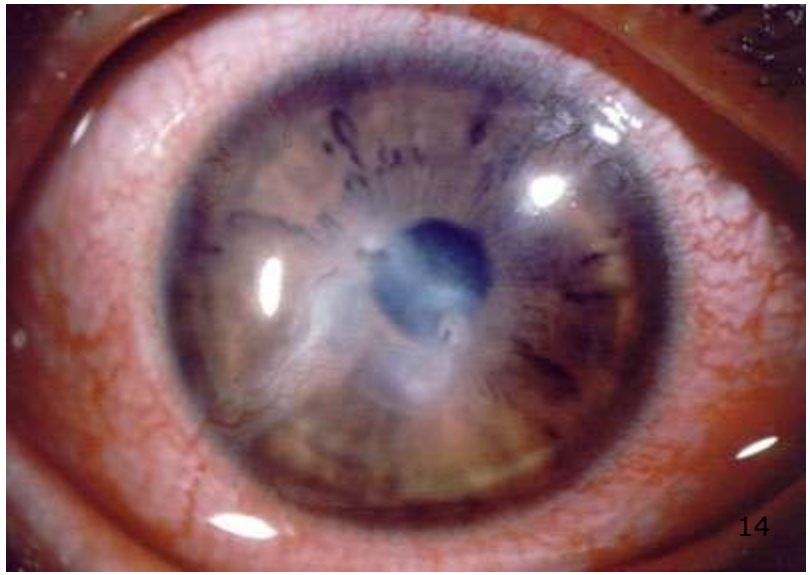

- Dietary restrictions that trigger reaction
  - well-known rosacea tripwires:
    - hot spices, alcohol, heated beverages
  - lesser-known rosacea tripwires:
    - chocolate, tomatoes, citrus fruits
  - No significant effect:
    - dairy products (milk, cheese)
- Environmental restrictions that may trigger reaction
  - Sun exposure, strong winds, strenuous exercise, change in humidity/weather
  - Emotional factors (stress, fear, anxiety, embarrassment)
  - Heliobacter pylori (bacteria linked with stomach ulcers)
  - Vasodilators

# Ocular Manifestations of Rosacea

- Meibomian gland dysfunction
- Blepharitis
- Hordeolum
- Chalazion



- Inflammation and infection of the lids
- Poor tear film
- Conjunctival injection
- PEE (SPK)
- Corneal vascularization
- Corneal thinning



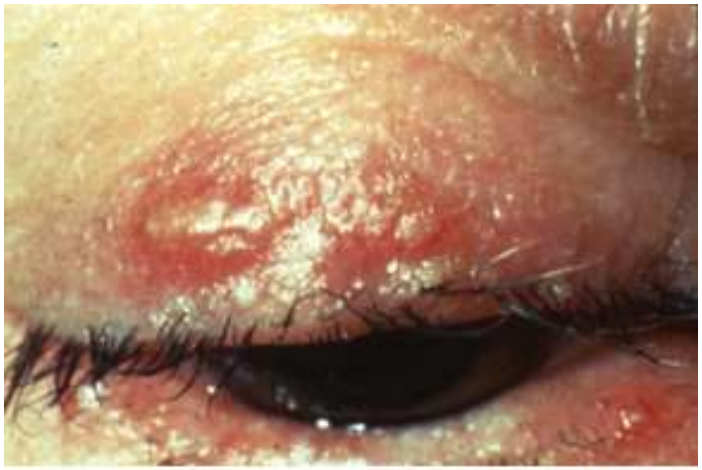
# Rosacea

- Papules and pustules
- Telangiectasias
- Staph blepharitis
- Madarosis
- Meibomian gland dysfunction
- Conjunctivitis



# Ocular Rosacea

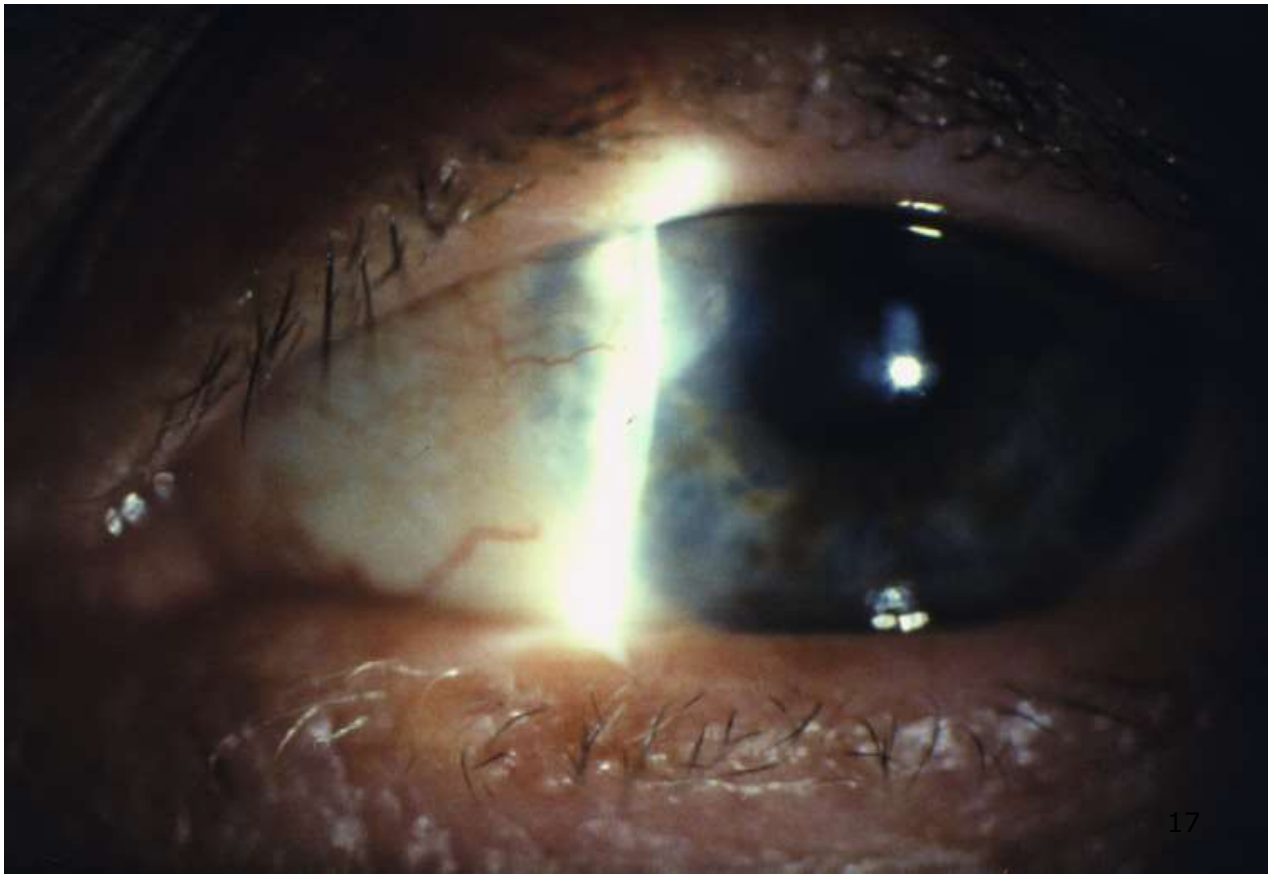
- **Infection and inflammation of the lids**





# Ocular Rosacea

- **Keratitis**
  - **Corneal vascularization and thinning**



# Ocular Rosacea

## Treatment

- **Meibomian gland dysfunction**
  - **Warm compresses, lid scrubs, gland expression**
- **Blepharitis**
  - **Same as staphylococcal blepharitis**
- **Keratitis**
  - **Artificial tears/ointments**
  - **Steroids if severe inflammation**



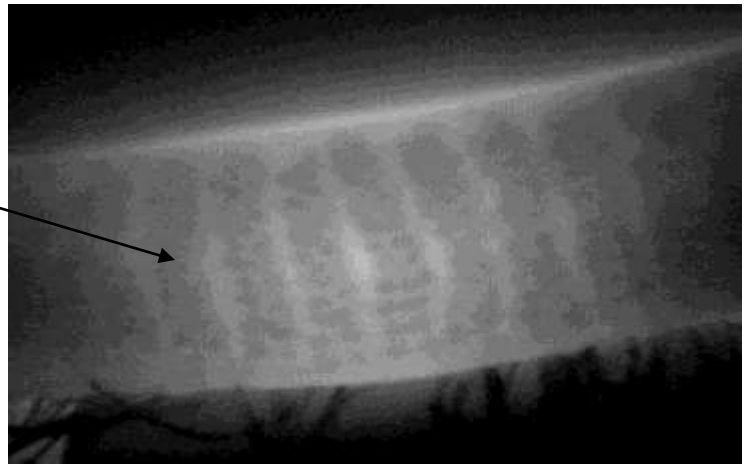
# Ocular Rosacea

- **Follow-up**
  - **Without corneal involvement - weeks**
  - **With corneal involvement - days**
- **Patient education**
  - **Lid hygiene**
  - **Dietary restrictions (triggers)**
  - **Environmental triggers**

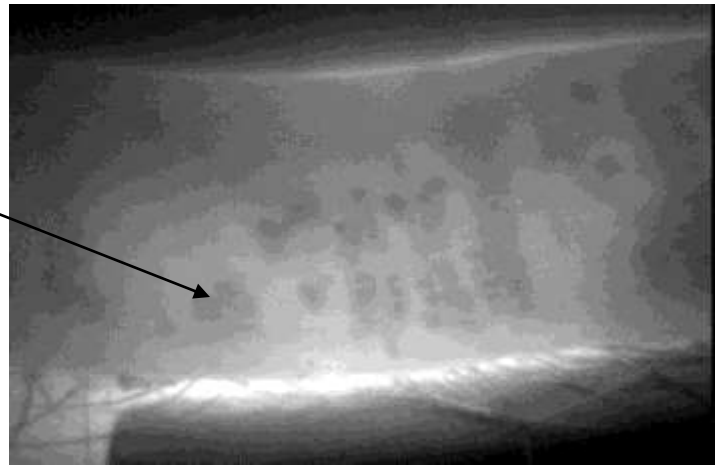


# Meibomian Gland Dysfunction

- **Normal meibomian glands**

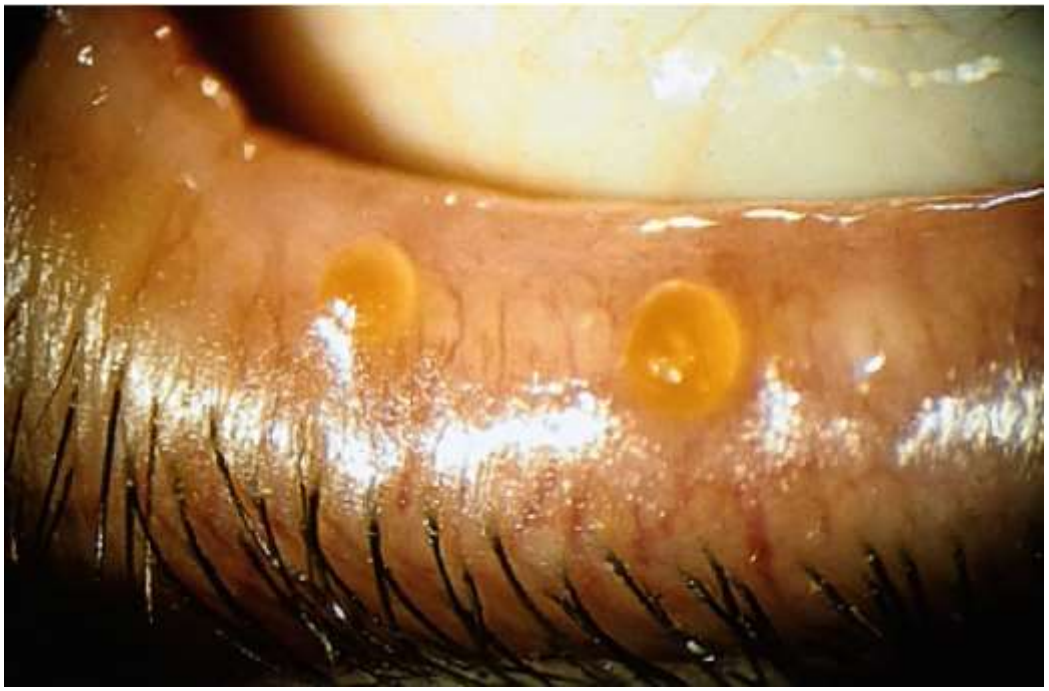


- **Abnormal meibomian glands**



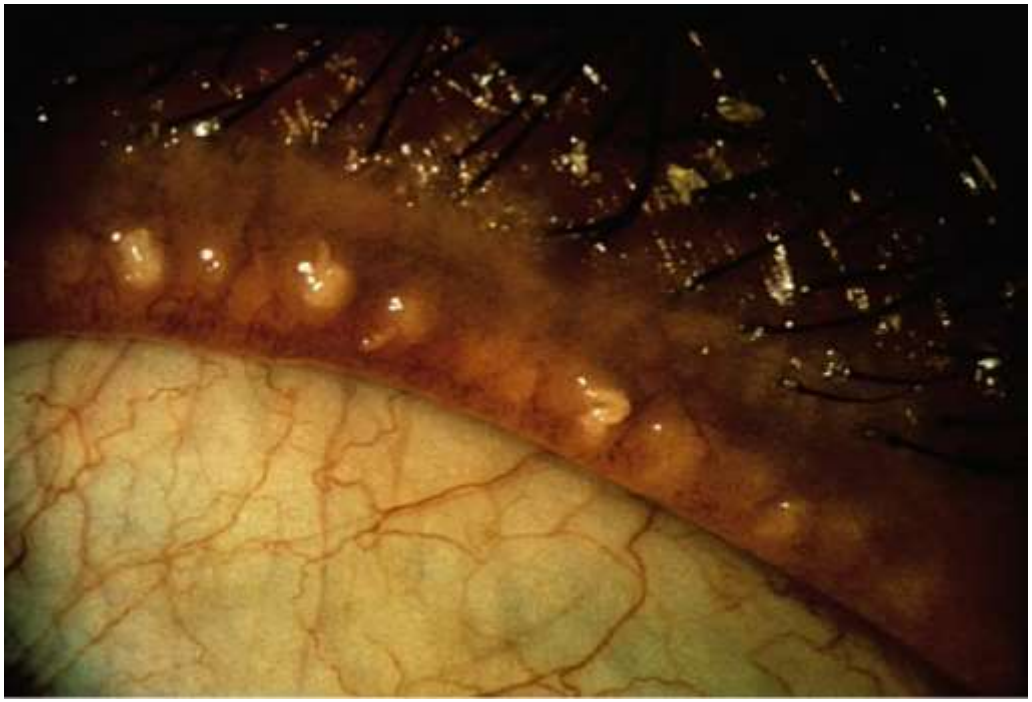
# Meibomian Gland Dysfunction

- Plugged meibomian glands
- Meibomian ‘caps’

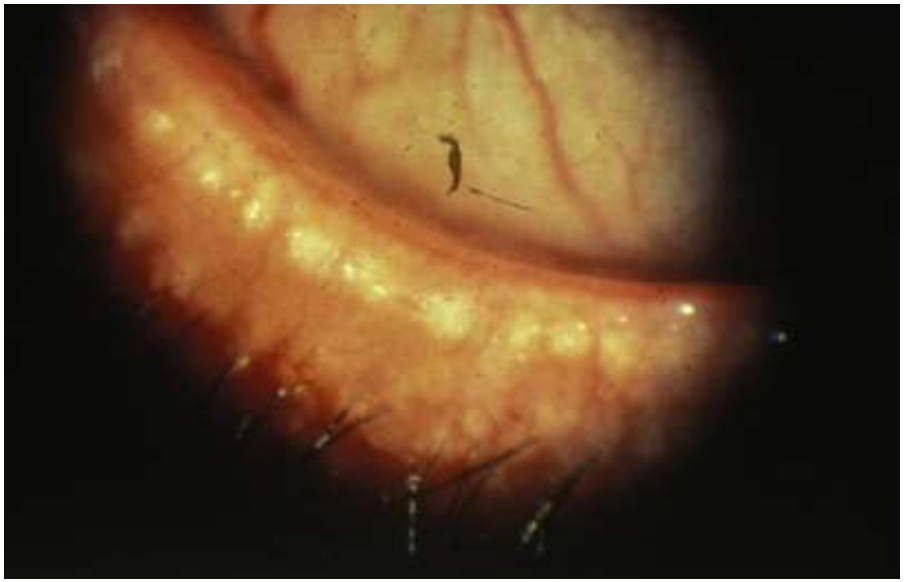


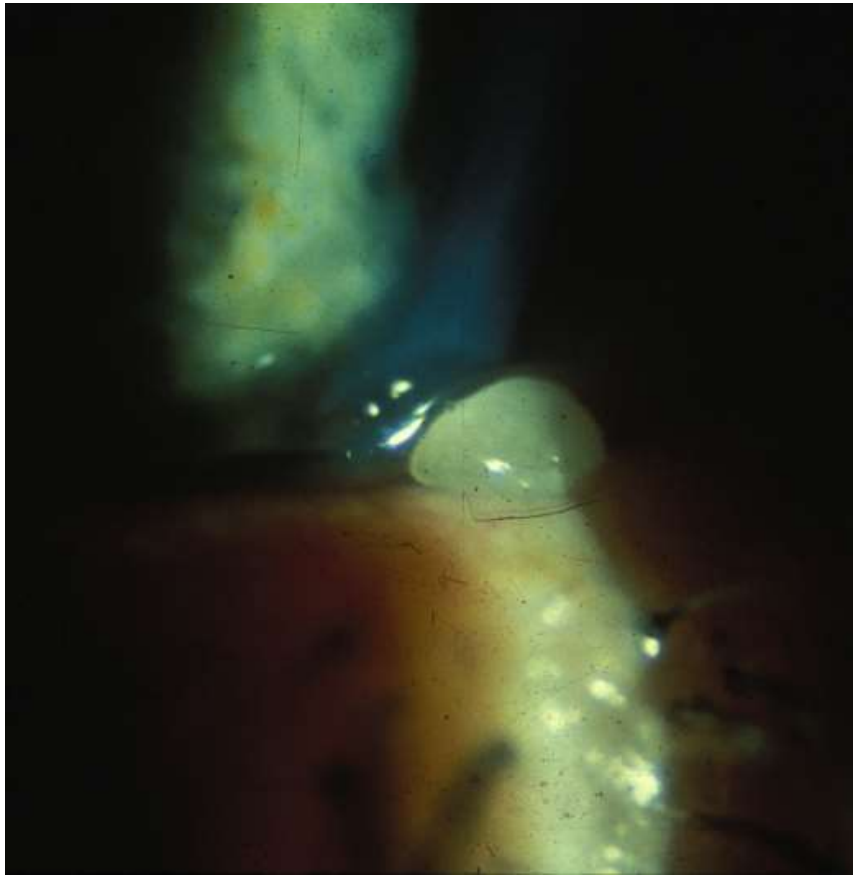
# Meibomian Gland Dysfunction

- Significant pouting of meibomian glands
- Inflammation of lid surrounding glands
- Serrated lid margin
- Tarsal conjunctival streaking
- Conjunctival injection



# Meibomian Gland Dysfunction



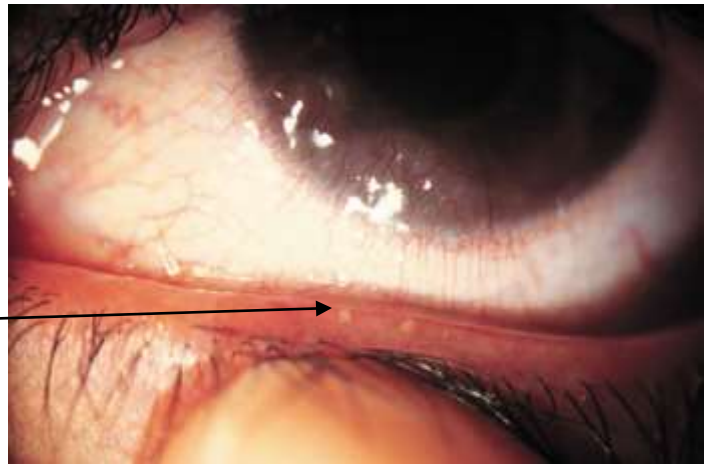


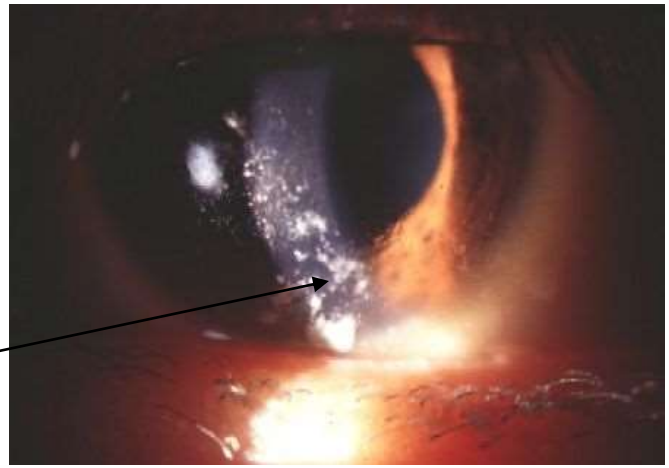
# Meibomian Gland Dysfunction

- **Diagnosis usually made clinically**

- **Forced expression of the meibomian glands**
  - **“Toothpaste” test - thickened secretions are forced out of some glands**
  - **Many glands will yield no visible secretion**

  - **Thickened secretions in the tears**





# Meibomian Gland Dysfunction

**Treatment**

- **Warm compresses**
  - **Softens any material on the meibomian gland orifices**
  - **Softens thickened secretions in glands**
- **In-office and at-home meibomian gland expression (digital massage)**
  - **Pressure on the lids from the part of the tarsus away from lid margin towards lid margin (opening)**
  - **Open the ducts of the glands and express the thickened secretions**

# Meibomian Gland Dysfunction



# Meibomian Gland Dysfunction

**Treatment**

- **If disease severe or not responding to therapy**
  - **Consider doxycycline 100 mg bid for 6 to 10 weeks**
  - **Taper to lowest dose or, if possible, discontinue**
  - **Works by inhibiting lipase production**



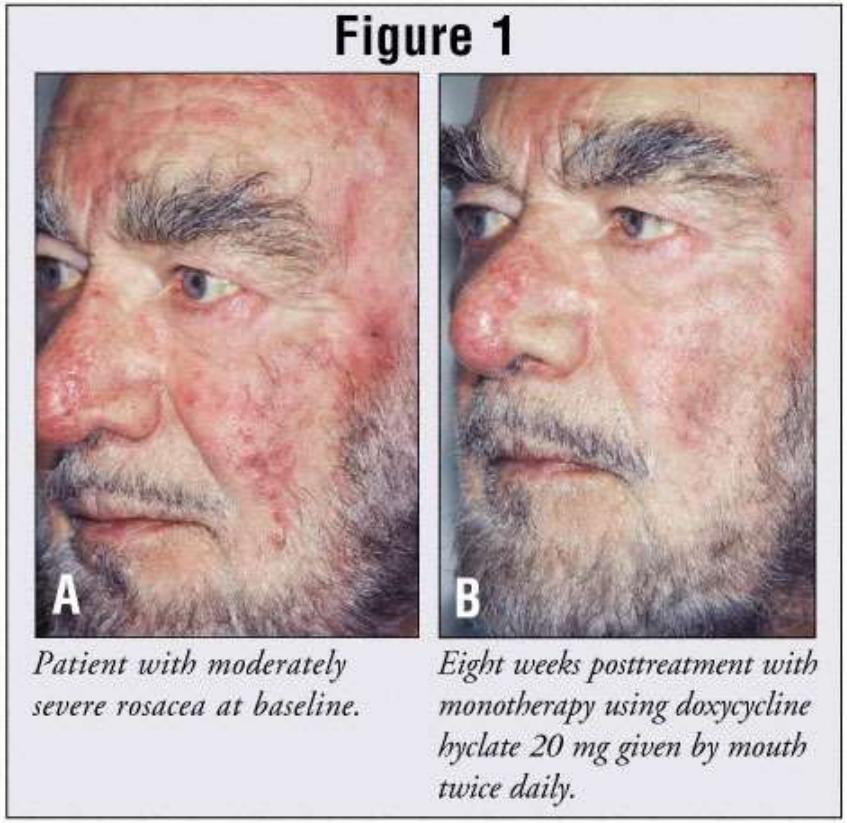
- **Manage dry eye**
- **Evaluate overlap with staphylococcal blepharitis**
  - **Treat both conditions**
- **Evaluate overlap with seborrheic blepharitis**
  - **Treat both conditions**
- **Evaluate if rosacea present**
  - **Consider dermatologic consult**



# Meibomian Gland Dysfunction

**Follow-up**

- **1 week**
- **Then 3-4 weeks later**

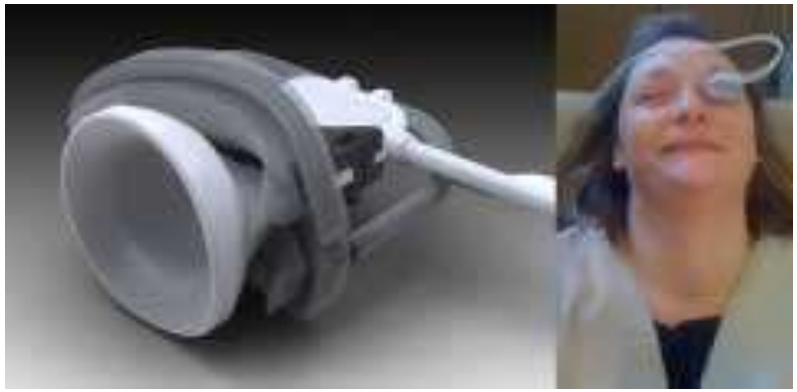



**Patient education**

- **Chronic condition**
- **No cure, management only**

# LipiFlow Thermal Pulsation System (TearScience)

- Meibomian expression in-office treatment device
- Invented by Donald Korb, O.D.
- Applies heat to tarsal side, pressure to skin side
- Can adjust heat/pressure; 12 minute sessions
- FDA-approved 7/11/11





# Intense Pulsed Light Therapy (IPL)



# Intense Pulsed Light Therapy (IPL)

- Non-laser, high intensity light
- Hand piece contacts skin
- Intensity, duration and wavelengths vary based on skin type/treatment goals
- Thermal heating of meibum, superficial telangiectasia




- Potential benefit for ocular rosacea and dry eye related to meibomian gland dysfunction (MGD)
- Increases lipid layer and noninvasive TBUT

# Impetigo

- Skin infection with *Staphylococcus aureus* or *Streptococcus pyogenes*
- Most common
  - Children under age 6
  - Sometimes very elderly
  - Exposed to poor hygienic conditions
- Contagious



# Impetigo

- Rash with crusts (‘honey crusts’)
- Face, distal extremities







# Impetigo

- Oral antibiotic
  - Culture and susceptibility testing
  - Cephalosporin
  - Macrolide
  - Dicloxacillin
- Wash involved areas
- Topical antibiotic after washing
  - Bacitracin
  - Mupirocin



# Erysipelas

- Skin infection involving the dermis
  - Superficial dermal lymphatics swell up
  - local redness, heat, swelling, and highly characteristic raised, indurated border
- Most common in young and elderly
- Legs and face most frequently affected sites
- Caused by *Streptococcus pyogenes*





# Erysipelas

- Fever, chills
- Malaise
- Painful edematous patches on skin
- If on face, can continue into lids
- Sharply demarcated areas of swelling and redness



# Erysipelas







- Treatment
  - PCP and/or dermatologic consult
  - Antibiotic: Penicillin G (injected) or penicillin V (PO)

# THANK YOU!